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Beyond Technological Solutionism

Sustainable Digitalization in High-Visitor Attractions

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Sustainable Digitalization in High-Visitor Attractions

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ABSTRACT

High-visitor attractions such as museums, botanical gardens, and heritage venues worldwide collectively receive over a billion visits annually, yet the sustainability of their daily operations remains underexplored. While digital tools are increasingly promoted as solutions to operational and environmental challenges, it remains unclear how such tools translate into contexts characterized by public-facing operations. Moreover, technological adoption is not inherently beneficial and may introduce new environmental, organizational, and ethical considerations.

This study explores how high-visitor attractions currently manage visitor flow and resource use in daily operations, and what this reveals about the conditions under which digital tools could support sustainable operations. An exploratory qualitative approach was adopted, based on semi-structured interviews with practitioners across six high-visitor attractions in Stockholm, Sweden. An analytical framework inspired by software requirements engineering was used to assess operational assumptions related to visitor predictability, data availability, staff capacity, resource availability, decision-making structure, and sustainability integration.

Findings show that visitor flow management relies primarily on staff experience and situational judgment, while digital data is used mainly retrospectively rather than as real-time operational support. Resource management practices are fragmented, sustainability is often treated as a strategic ambition rather than an operational driver, and no venue fully satisfied the conditions required for idealized digital tool deployment.

The study identifies structural institutional barriers, fragmented authority, and tensions between heritage preservation and resource efficiency as key factors shaping sustainable digital adoption. It concludes that responsible digitalization in high-visitor attractions depends less on technological capability and more on the organizational and institutional conditions in which digital tools are implemented. Rather than positioning digitalization as a universal solution, the study emphasizes the importance of understanding operational realities before introducing digital tools into complex visitor environments.

SAMMANFATTNING

Välbesökta attraktioner såsom museer, botaniska trädgårdar och kulturarvsinstitutioner tar tillsammans emot över en miljard besök årligen, men hållbarheten i deras dagliga verksamhet är fortfarande relativt outforskad. Samtidigt lyfts digitala verktyg i allt högre grad fram som lösningar på operativa och miljömässiga utmaningar, men det är fortfarande oklart hur dessa verktyg fungerar i kontexter som präglas av publik verksamhet. Teknologisk implementering är inte heller per automatik gynnsam och kan medföra nya miljömässiga, organisatoriska och etiska konsekvenser.

Denna studie undersöker hur välbesökta attraktioner i dagsläget hanterar besöksflöden och resursanvändning i den dagliga verksamheten, samt under vilka förutsättningar som digitala verktyg skulle kunna stödja en mer hållbar verksamhet. En explorativ kvalitativ metod har använts, baserad på semistrukturerade intervjuer med anställda från sex välbesökta attraktioner i Stockholm, Sverige. Ett analytiskt ramverk inspirerat av kravhantering inom mjukvaruutveckling användes för att analysera operativa antaganden kopplade till besöksförutsägbarhet, datatillgång, personalens kapacitet, resursförutsättningar, beslutsstruktur samt integrering av hållbarhet.

Resultaten visar att hantering av besöksflöden i huvudsak bygger på personalens erfarenhet och situationsbaserade bedömningar, medan digitala data främst används retrospektivt snarare än som operativt stöd i realtid. Resurshanteringen är fragmenterad, och hållbarhet framstår ofta som en strategisk ambition snarare än en operativ drivkraft, samtidigt som inget av de studerade fallen fullt ut uppfyller de villkor som krävs för idealiserad implementering av digitala verktyg.

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Studien identifierar strukturella och institutionella hinder, fragmenterat beslutsfattande samt spänningar mellan kulturarvsbevarande och resurseffektivitet som centrala faktorer som påverkar hållbar digitalisering. Avslutningsvis konstateras att ansvarsfull digitalisering i välbesökta attraktioner i mindre grad handlar om teknisk kapacitet i sig, och i större grad om de organisatoriska och institutionella förutsättningar inom vilka digitala verktyg implementeras. Studien argumenterar därför att digitalisering inte ska ses som en universallösning, och betonar vikten av att förstå operativa realiteter innan digitala verktyg införs i komplexa besöksmiljöer.

Keywords: sustainable digitalization, high-visitor attractions, socio-technical systems, operational planning, visitor flow, resource management

Nyckelord: hållbar digitalisering, välbesökta besöksattraktioner, socio-tekniska system, operativ planering, besöksflödeshantering, resurshantering

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LIST OF ACRONYMS

HVA	High-Visitor Attractions
CDR	Corporate Digital Responsibility
SDG	Sustainable Development Goal
GDPR	General Data Protection Regulation
HM	History Museum (anonymous)
IM	Interactive Museum (anonymous)
NRM	Naturhistoriska Riksmuseet
SKM	Scenkonstmuseet
EAV	Edvard Andersons Växthus
VM	Vasamuseet
SFV	Statens Fastighetsverk
IoT	Internet of Things
AI	Artificial Intelligence
VR	Virtual Reality
AR	Augmented Reality

1 INTRODUCTION

Museums, theme parks, and other visitor attractions are familiar spaces to most people; it is estimated that more than a billion people visit museums worldwide every year [1]. These environments are often characterized by high visitor volumes, complex logistics, and intensive day-to-day operations. As a result, two interconnected aspects become particularly relevant; firstly, the need for these attractions to operate in environmentally sustainable ways. Visitor-intensive environments can involve substantial energy and water consumption, with operational impacts strongly linked to visitor numbers and facility scale [2]. Secondly, their potential to generate broader impact due to their large reach and influence. Due to the high number of visitors they accommodate, even small operational decisions may have significant cumulative environmental and organizational effects. In addition to their operational footprint, high-visitor attractions are visible public environments, and their everyday operations can influence how visitors perceive sustainability.

High-visitor attractions rely on continuous operational planning to manage visitor flows, staffing, energy use, maintenance, and opening hours, while simultaneously delivering positive visitor experiences and responding to increasing environmental sustainability pressures, reflected in the 17 Sustainable Development Goals (SDGs), which call for urgent action on climate change and environmental protection. These global commitments are reshaping expectations for how publicly funded institutions manage operations and address environmental impact, particularly in relation to SDGs 12: Responsible Consumption and Production and SDG 13: Climate Action [3]. Many of these operational decisions are shaped by a combination of internal constraints and external factors such as weather conditions, seasonal variation, and unexpected disruptions. Operating sustainably within this context is therefore not only a strategic ambition, but an everyday practical challenge embedded in routine decision-making.

In parallel, digital technologies have advanced rapidly over the past decade, becoming deeply integrated into both everyday life and specialized domains such as industry, logistics, and scientific research. Tools for data collection, simulation, and predictive analysis are increasingly used to support planning and optimization in complex systems [4]. This raises the question of whether such technologies can be meaningfully applied beyond traditionally technical domains and into sectors that are more publicly oriented, such as museums and recreational venues, contexts where digital infrastructure exists, but is often less visible to visitors.

Digitalization is frequently presented as a single problem solver for efficiency and accessibility challenges. Although digitalization can be beneficial to achieve environmental sustainability efforts, technological adoption is not inherently beneficial and may introduce new environmental, organizational, and ethical considerations. Responsible digitalization requires an understanding of when technology meaningfully supports sustainability goals, and when simpler, non-digital approaches may be more appropriate in order to avoid unintended outcomes that may undermine, rather than support, sustainability ambitions [5, 6]. In high-visitor environments, where operational complexity is already high, this balance becomes particularly delicate.

Sustainability-related challenges in such environments are also increasingly visible not only from an operational perspective, but also from the visitor experience itself, where inefficiencies or resource-intensive practices may be directly observed. This visibility highlights the importance of critically reflecting on how these environments are managed and where improvements may be needed.

Understanding how High Visitor Attractions (HVAs) currently operate is therefore a necessary precondition for any meaningful assessment of digital tools in this context. Without this empirical foundation, proposals for technological solutions risk assuming conditions that do not exist, introducing tools into environments that are not prepared for them, and generating the unintended consequences that Corporate Digital Responsibility (CDR) warns against. This thesis is motivated by that gap: before asking which digital tools could improve sustainability in HVAs, it is necessary to ask whether the conditions for sustainable introduction exist at all. This question sits within a broader concern about how sustainable digitalization can be pursued in complex, public-oriented institutions, and how CDR and sustainability principles can be operationalized in practice rather than applied only at a policy level.

This thesis, therefore, adopts an exploratory approach focused on the everyday operational context of high-visitor attractions, with particular attention to museums due to their accessibility and relevance within the study. Rather than evaluating or proposing a specific technological solution, the aim is to build an empirically grounded understanding of current practices and challenges. The focus is placed in particular on visitor flow management (movement and distribution of visitors within and into the venue) and resource use (most often energy, water, waste, and staffing), as these represent operational areas where sustainability considerations and operational decision-making are most directly observable, and where environmental and organizational impacts become particularly visible in practice.

Therefore, this thesis aims to answer:

"How do high-visitor attractions currently manage visitor flow and resource use in daily operations, and what does this reveal about the conditions under which digital tools could support sustainable daily operations?"

By addressing this question, the thesis seeks to contribute to a better understanding of operational sustainability within high-visitor environments and to explore how digitalization can be approached responsibly in practice. Rather than assuming that more technology necessarily leads to better outcomes, the study emphasizes the importance of understanding existing operational realities, organizational needs, and sustainability considerations before introducing digital tools into complex visitor environments.

2 BACKGROUND

To situate this study, the following section reviews three areas of literature, providing conceptual and contextual background. Section 2.1 introduces high-visitor attractions as an operational and research context. Section 2.2 examines Corporate Digital Responsibility and the risks of inadequately implementing digitalization. Section 2.3 discusses software requirements engineering as the conceptual basis for the analytical framework developed in this thesis. Together, these sections establish the foundation for the assumption-based analysis that follows.

2.1 High-Visitor Attractions Overview

High-visitor attractions, which will be referred to as "venues" or "HVAs" interchangeably, are vital for any country's tourism and entertainment industry, often working as the *Raison d'être* for many tourists, meaning that this type of attractions is the major reason for visiting a destination [7-9].

High-visitor attractions are often seen as merely educational or for entertainment. Most people have visited one of these attractions with that purpose, but rarely stop to think about what has to happen day-to-day to ensure that these venues can operate effectively. Despite the proven relevance to a country's tourism sector, visitor attraction still remains an area that is under-researched [10-12].

First and foremost, the term for these venues must be defined. As repeatedly defined by the English Tourism Council, high-visitor attractions refers to "...a permanently established excursion destination, a primary purpose of which is to allow public access for entertainment, interest or education; rather than being primarily a retail outlet or a venue for sporting, theatrical or film performances […] capable of attracting day visitors or tourists as well as local residents" [13]. These attractions further divide into different categories and types, as observed in *Table 1*. Some examples are museums, animal attractions, and theme parks.

Table 1: Categories and Types of Visitor Attractions (adapted from Dewhurst, 1996 as presented in [10])

Attraction category	Attraction types	Constituent attractions
Historico-cultural	Religious sites	Abbeys, cathedrals, chapels, priories

Attraction category	Attraction types	Constituent attractions
	Museums and galleries	Art galleries, open air museums, traditional museums, science centers
	Historic sites	Castles, landmarks, monuments, palaces
	Interpretative heritage sites	Interpretative centers, heritage sites
	Multi-faceted historic sites	Castles, docklands, historic houses, palaces
Environmental	Animal attractions	Safari parks, wildlife parks, zoos, rare breed farms, nature centers, aquaria
	Parks and gardens	Botanic gardens, outdoor activity parks, public parks
	Country parks	Country parks, reservoirs
Entertainment	Leisure and recreation complexes	Leisure centers, leisure pools, recreation centers, water parks
	Amusement parks	Pleasure beaches, pleasure parks
	Theme parks	Indoor parks, outdoor parks, beach resorts
	Themed retail outlets	Antique centers, garden centers, retail, and leisure parks
	Workplace industrial visit centers	Craft workshops, factory shops
Miscellaneous other		Arboretums, piers, themed transport

HVAs daily activities rely on operational planning and management. These operations typically span several interconnected domains, including visitor flow and crowd management, queue management, resource-related operations such as energy, water, and waste use, staffing, safety and security, maintenance, and revenue-related activities. Together, these domains shape how visitor experiences are delivered and how operational resources are allocated on a daily basis.

When visitor attractions were first established, most relied on manual processes and experience-based decision-making, often unprepared for high crowds and unpredictable events. In the 90s to 2000s a more formal approach to operations management was adapted, texts such as *Heritage Visitor Attractions: An Operations Management Perspective* (Leask & Yeoman, 1999), and *Management in Museums* (Moore, 1999) began to introduce terms such as, quality control, and capacity planning, as well as, applying concepts like leadership, organizational change theory, and quality management to the visitor attraction sector. As time passed and crowds grew, adaptation and development of visitor attractions' management operations did so as well.

The COVID-19 pandemic accelerated digital transformation in visitor attractions; nowadays, it is common for these places to use timed-entry ticketing, online booking, and digital guides or maps, becoming the standard rather than exceptional [14]. At the same time, there has been an increase in the use of real-time visitor analytics, data-driven

management systems using predictive and prescriptive analytics, immersive technology, among other technologies, often making use of AI, IoT, VR, AR, and machine learning, this with the goal of increasing innovation, accessibility, and optimize operations [15, 16]. The level of adoption for digital technologies in visitor attractions is limited by high cost and organizational resistance to change, due to lack of sufficient financial and technological capacities [17].

Although there was a high interest in developing strategic and visitor centered operations, environmental operations, such as water, energy, and waste, have received less attention. Moreover, advances in these practices have primarily been through environmental policy pressures rather than operational optimization [18].

Technology and sustainable policy development, as well as, changing patterns in tourism and visitor demand, continue to place pressure on visitor attraction management operations. “[S]urvival in the future will depend on effective management responses to current and future challenges” [10].

This historical development highlights how visitor attractions have progressively become more data-driven and technologically supported in their operational planning. However, the role of digital systems in supporting operational sustainability remains less explored.

2.2 Digital Responsibility

As organizations continue to face complex environments, digitalization has become an increasing response to them. Often recurring to digital tools to address customer experience, service quality, and productivity [19]. As previously mentioned, digitalization continues to make deep changes in both the private and professional lives of people; the further they develop, technologies also create attractive opportunities for businesses in various sectors [20].

However, digitalization is not inherently beneficial and can in turn cause the opposite effect of what is intended, more specifically on social, economic, and environmental landscapes. While “digital technologies grant documented operational and economic benefits, they can also activate unwanted consequences for the environment and society” [21]. There are three severe risks when implementing digitalization inadequately:

- **Rebound effects:** “improvements in efficiency often lead to cost reductions that provide the possibility to buy more of the improved product or other products or services” [22]. For example, smart crowd management systems optimize visitor flow, reducing congestion but improved experience encourages repeat visits and longer stays, increasing total resource use (lighting, climate control, cleaning).

- **Shifted environmental burdens:** “a consequence that can occur when efforts to minimize a problem in one context have unintended negative impacts on another” [23]. For example, a venue shifts from printed brochures to digital apps and interactive guides reduces paper waste locally but increases server energy use, cloud storage emissions, and device charging infrastructure demand (e.g. phone charging).

- **Increased organizational complexity:** when digitalization is introduced it will increase complexity since it requires maintenance, software updates, cybersecurity management, and staff training; adding an extra layer of complexity. Rather than simplifying operations, they can introduce new dependencies and failure dependence vulnerabilities (where one failure can affect multiple connected parts of the system) [24]. For example, introducing real-time visitor analytics dashboards may require coordination with external system providers for development and maintenance, staff training for data interpretation, collaboration between IT and operational teams, and continuous adaptation of operational decisions based on live inputs.

More formally known as Corporate Digital Responsibility (CDR), a consensus definition for this term has not yet been established. Here Herden *et al.’s* definition will be used “[...]an extension of a firm’s responsibilities which takes into account the ethical opportunities and challenges of digitalization” [20]. While Herden et al.'s definition foregrounds the ethical dimension, this study interprets CDR more broadly to encompass the social and environmental consequences of digital adoption as well; grounded in the understanding that ethical, social, and environmental impacts are deeply

interconnected in the context of publicly funded institutions. CDR extends traditional corporate social responsibility into the digital domain, addressing the ethical, social, and environmental dimensions of technology adoption. It is crucial that digital tools are implemented responsibly, carrying out a careful and context-aware adoption, while considering sustainable impact, shifting away from the wrong belief that digitalization is inherently beneficial.

In high-visitor attractions, Corporate Digital Responsibility is particularly relevant since digital tools intended to support sustainability may also introduce new environmental and organizational impacts. For example, digital visitor management systems can reduce material use but increase reliance on energy-intensive infrastructure and external service providers, while efficiency improvements in visitor flow may contribute to rebound effects through increased capacity or usage. These examples illustrate that the sustainability outcomes of digitalization are context-dependent and must be assessed in relation to operational realities and sustainability objectives, rather than assumed to be universally positive.

2.3 Software Requirements and Assumptions

Digital tools are typically designed and evaluated through requirements engineering approaches, which define the conditions under which a system should operate. These requirements are commonly divided into functional requirements, which describe what a system should do, and non-functional requirements, which define how the system should perform in terms of quality attributes such as performance, usability, and reliability [25,26].

However, in socio-technical environments, system performance is not determined solely by technical specifications. Instead, it depends on organizational capacity, data availability, user behaviour, and contextual constraints. As a result, digital systems implicitly rely on a set of operational assumptions about the environment in which they are deployed.

Sustainability has historically received limited attention in software engineering, and is described as “a concept not often addressed within software engineering from a software requirements perspective” [27]. This reflects a broader gap in software engineering practice, where environmental considerations are not systematically included in requirements definition and prioritization. To address this, it has been argued that environmental sustainability should be explicitly included as a non-functional requirement to support a more holistic understanding of system impacts [27, 28].

In this thesis, these concepts are adapted into a set of Analytical Assumptions that represent idealized conditions under which digital tools for visitor flow and resource management would function effectively in high-visitor attractions. In this case, six Assumptions were identified, which are described in Section 5.1.

3 METHOD

The study was initially considered as an evaluation of whether digital technologies commonly applied in industrial and data-driven sectors (e.g., digital twins, industrial IoT systems, AI-based predictive analytics) could be transferred to visitor-driven environments such as high-visitor attractions. However, this direction was refined during the early stages of the research design, as it became evident that limited empirical work exists on operational conditions and decision-making processes in these contexts. As a result, the study adopts an exploratory qualitative research design focused on understanding current practices before assessing the role of digital tools.

Rather than evaluating a specific technology or system, the study focuses on understanding current practices, challenges, and perspectives among practitioners working in visitor attractions. This allows the research to identify whether there is a perceived need for digital tools or approaches to support sustainable operations.

The research, therefore, focuses on gathering practitioner insights through interviews with individuals involved in the operations and management of high-visitor cultural institutions, such as museums and similar visitor venues.

3.1 Context and Participants

The study focuses on high-visitor attractions, particularly museums and similar visitor environments where daily operations involve managing visitor flows, facilities, and resource use. These environments are characterized by fluctuating visitor numbers, event-based programming, and operational decisions that must balance visitor experience, safety, and resource efficiency.

Participants were recruited from organizations in Sweden, particularly Stockholm, involved in visitor attraction operations, visitor management, or sustainability-related roles. Individuals were identified through professional contacts, institutional websites, and publicly available organizational information. A spreadsheet was used to track progress in communication with each organization and its designated contact person.

The study aims to collect insights from practitioners with operational knowledge, and the goal is exploratory understanding rather than statistical generalization.

The study included a total of seven participants from high-visitor attractions in Stockholm, Sweden. Two participants contributed in the exploratory pre-study phase, one of which was later re-interviewed as part of the main data collection in order to provide supplementary insights relevant to the final analysis. The remaining participants contributed through semi-structured interviews, forming the primary empirical material of the study.

Participants held roles related to operations, exhibitions, sustainability, or visitor management. The majority of participants represented museums, reflecting the accessibility of these institutions during data collection. One participant represented a greenhouse located within a botanical garden. Below (*Figure 1*), participants are listed together with the part of the study to which they contributed. Three museums chose to remain anonymous due to participant preference, institutional PR policies, or lack of formal approval for identification. To preserve anonymity while supporting reader understanding, these museums are described only through broad institutional categories, including an art museum, a history museum, and an interactive museum.

Venue	Role	Contribution
Art Museum (AM) *Anonymized*	N/A *Anonymized*	Pre-study (questionnaire)
History Museum (HM) *Anonymized*	N/A *Anonymized*	Interview
Interactive Museum (IM) *Anonymized*	N/A *Anonymized*	Interview
Naturhistoriska Riksmuseet (NRM) *Natural History Museum*	Head of Exhibitions and Learning	Pre-study (interview) & Interview
Scenkonstmuseet (SKM) *Performing Arts Museum*	Exhibitions Manager	Interview
Vasamuseet (VM) *Vasa Museum*	Sustainability Strategist	Interview
Edvard Andersons Växthus (EAV) *Edvard Andersons Conservatory (Greenhouse) part of Bergianska Botanical Garden*	Operations Staff	Interview

Figure 1: Study Participants

3.2 Data Collection

All data collection was done through the contact person at the corresponding visitor attraction, primarily through interviews. The only case in which a questionnaire was used instead of an interview was with one participant involved in the pre-study phase (AM). Interviews were conducted online (three) or in person (three), depending on participant availability and preference, and were audio recorded with participant consent to support accurate transcription and analysis.

An initial pre-study through interviews and questionnaires was conducted to refine the focus of the thesis and evaluate whether the proposed questions were appropriate for addressing the research question prior to more extensive data collection. The pre-study helped narrow the scope of the study from general operational management practices to a more focused examination of visitor flow and resource use in high-visitor attractions.

3.3 Question Design

The pre-study questions were designed to explore how operational planning and sustainability considerations are addressed in practice within high-visitor environments. Based on the results of this initial phase, the main interview questions were refined to address identified gaps and to capture empirical evidence related to three operational domains: visitor flow, resource management, and sustainability.

In addition, the interview design aimed to generate data relevant to six Analytical Assumptions identified, established between the pre-study and the main interviews. These Assumptions are further explained in Section 5.1. This dual purpose influenced both the content and sequencing of the interview questions.

A semi-structured interview format was selected due to the exploratory nature of the study, allowing for both flexibility and consistency across interviews. This approach enabled participants to elaborate on aspects most relevant to their organizational context while ensuring that comparable themes were addressed throughout the data collection process. Questions were intentionally kept open-ended and were not explicitly framed around the Assumption Framework, allowing assumption-related evidence to emerge naturally from participants' own descriptions rather than through direct prompting. Follow-up probes were prepared for each thematic area and used when additional clarification or detail was required.

The interview guide was organized into six thematic sections, progressing from participant role and organizational context to visitor flow, resource management and sustainability, digital tools, digital tool suitability and responsible adoption, and concluding reflections.

3.4 Data Analysis

The collected data was analyzed using deductive thematic analysis to identify recurring operational patterns and challenges across the studied venues [29]. In a second stage of analysis, findings were interpreted using the six Analytical Assumptions introduced in the background section and further defined in Section 5.1. This enabled the study to assess how current operational conditions align with the requirements commonly associated with digital planning tools.

3.5 Ethical Considerations

Participation in the study was voluntary, and participants were informed about the purpose of the research before contributing. Interviewees were asked for consent before participation, including consent for audio recording where applicable. Recordings were used solely to support the transcription process and facilitate analysis.

All data was handled confidentially, and identifying information about participants or organizations was included only when explicit permission was provided. Interview materials, including recordings and transcripts, will not be shared outside the context of this research without the explicit consent of participants.

3.6 Limitations

This study is exploratory in nature and therefore focuses on gathering qualitative insights rather than producing statistically generalizable results. Data collection was influenced by practical constraints, including the timeframe and organizational accessibility. As outreach was conducted independently by the researcher, access to potential participants depended on publicly available contact information and voluntary responses, which may have limited access to certain organizations or roles within institutions.

Participation depended on the availability and willingness of professionals working in visitor attractions, which limited the number of potential respondents. Additionally, responses reflect the perspectives and experiences of individual participants. The level of detail provided may vary depending on participants' roles and responsibilities within their respective organizations.

4 RESULTS

This section presents empirical findings from six high-visitor attractions in Stockholm, gathered from interviews. Art Museum (pre-study questionnaire) is excluded from this section due to the shallower data format, as noted in Section 3. It addresses the first part of the research question, *how High-Visitor Attractions (HVAs) currently manage visitor flow and resource use in daily operations*, and provides the empirical basis from which the analysis in Section 5 is drawn. Findings are organized into four subsections: visitor flow management, resource management, sustainability in operational planning, and digital tools currently in use.

4.1 Visitor Flow Management in Practice

Across all six venues, visitor flow was managed primarily through staff presence and real-time judgment rather than digital planning tools. Nevertheless, the nature and scale of flow challenges varied considerably depending on venue size and visitor profile.

At larger venues, peak periods were operationally intense. Naturhistoriska Riksmuseet (NRM) handles up to 35,000 visitors in a single school break week, compared to 600–700 on regular weekdays, creating situations where infrastructure, such as ventilation and catering, could not cope. As the Head of Exhibitions and Learning noted, *"the fans in the building can't cope with it. Complaints about noise, temperature […] that is not within our control."* The Interactive Museum (IM) similarly described peak days of 2,000–3,000 visitors against a typical 500 visitors, with a recurring bottleneck at the information desk. Smaller venues, such as the History Museum and the Edvard Andersons Växthus, reported fewer challenges, supported in part by the required booking for groups and lower overall volumes.

Most venues collected visitor count data through ticketing systems or sensors, though this was used mainly for retrospective analysis and budget forecasting rather than live operational decisions. Furthermore, experience and contextual knowledge shaped most in-the-moment responses across all venues, often alongside and sometimes in place of formal data. As described by the interviewee at Interactive Museum *"During peak and daytime, there's a lot based on experience and what worked or not previously. Both, but in the moment, experience."*

4.2 Resource Management in Practice

Resource management was characterized by fragmented responsibility across all venues studied. Energy was the most consistently monitored resource, though in several cases this monitoring was carried out by external parties rather than the organizations themselves. At Naturhistoriska Riksmuseet, energy is managed by the state property agency Statens Fastighetsverk under a contractual arrangement, limiting the museum's direct visibility into its own consumption. As one participant described it, *"energy is monitored by our landlord, Statens Fastighetsverk. It's a contractual thing."*

The History Museum reports monthly energy figures to the government. Additionally, Scenkonstmuseet noted that heritage building restrictions, such as the prohibition on solar panels, further constrained available options for energy reduction.

Edvard Andersons Växthus described the use of an automated environmental control system regulating temperature, ventilation, and lighting conditions within exhibition spaces, used to maintain stable conditions but also as a monitoring and regulation mechanism.

Materials and waste management were handled in various forms. Several venues described active efforts around the reuse and recycling of exhibition components: Naturhistoriska Riksmuseet (NRM) with strict demolition contracts, the History Museum through a materials exchange network across seven institutions, and Scenkonstmuseet through a dedicated circular materials store. At NRM, this approach was described as particularly rigorous: *"we are quite strict with how we recycle materials; we even check the receipts and confirmation."* Although some venues handle material and waste management more formally, tracking the sustainability credentials of externally sourced materials remained a recurring challenge.

4.3 Sustainability in Operational Planning

Sustainability was present as an organizational value across all venues, though its integration into daily operational decisions was uneven. Three venues: the Interactive Museum (IM), Naturhistoriska Riksmuseet, and Scenkonstmuseet, described formal structures that brought sustainability into routine practice, through dedicated coordinator roles, contractual obligations, and circular materials approaches, respectively. At Vasamuseet, sustainability was the most formally systematized in the dataset, the authority holds ISO 14001 certification since 2009 and conducts annual environmental reporting across all seven museums under its governance. Nevertheless, the sustainability specialist described a strategic shift underway, with the authority now directing attention toward social sustainability following progress on environmental targets.

At other venues, sustainability competed with other priorities. The History Museum offered the most direct account of this tension, with the participant noting that economic sustainability and visitor attraction currently take precedence in institutional discussions: *"economy and visitor experience are higher priority in how we discuss it. The most important thing is to attract visitors."* Moreover, at Naturhistoriska Riksmuseet, sustainability certification processes were described as quite administrative, suggesting a gap between formal commitments and their practical influence on daily decisions. However, these also serve as an opportunity to reflect and be more actionable when it comes to sustainability. As one participant described, *"those kind of like standards or certificates are quite blunt, to say, they're more administrative in a way. But maybe they act like the tags in our activity plan system. [...] they force us to reflect and adjust. And I think that might be a good thing."*

At the Edvard Andersons Växthus, sustainability improvements, including the replacement of oil-based heating, electrification of gardening equipment, and the introduction of bicycles in place of petrol vehicles, had been implemented at management level. However, the front-line participant described these as decisions made by others. This suggests that sustainability improvements were present but not operationally embedded at the level of daily front-line practice.

Finally, the environmental impact of digital tools themselves was raised at only two venues, suggesting limited awareness of this dimension of sustainability within day-to-day operational contexts.

4.4 Digital Tools Currently in Use

All venues used digital tools for planning and coordination, though no venue described a single integrated system covering visitor flow, resource management, and sustainability together. General-purpose platforms dominated: Microsoft Office was in use at most venues, Quinyx for workforce scheduling at Interactive Museum and Naturhistoriska Riksmuseet, and various ticketing systems depending on the venue. NRM additionally used ViaMetric for sensor-based visitor counting, while the History Museum described artificial intelligence as quite integrated into its operations, for visitor surveys, collection text recognition, and internal summaries, within a carefully managed legal framework.

Nevertheless, tools across all venues operated in parallel without a shared data layer between them. At Naturhistoriska Riksmuseet, Excel, a web-based task management tool, and Notion were all in simultaneous use for different functions. The History Museum faced the most constrained adoption environment, where all tools require government authority approval,

GDPR verification, and legal review before use, making adoption slow by design. As the practitioner acknowledged, *"it is quite a process to start using one tool. We can't shift so easily."*

5 ANALYSIS

Having established the empirical picture of how High-Visitor Attractions (HVAs) currently operate, this section turns to the second part of the research question: *what does current practice reveal about the conditions under which digital tools could support sustainable operations?* To answer this, the Assumption Framework introduced in Section 2.3 is applied to the empirical findings. Section 5.1 defines the framework and rating scale. Section 5.2 presents the Assumption Matrix. Section 5.3 analyzes each assumption in turn. Section 5.4 synthesizes the findings in relation to the research question.

5.1 Assumption Framework

As discussed in the background section, digital tools are typically designed and evaluated according to functional and non-functional requirements that define the conditions under which a system is expected to operate effectively. In software engineering, these requirements implicitly assume certain operational conditions, organizational capacities, and system environments. This thesis adapts these concepts into a set of Analytical Assumptions representing idealized conditions under which digital tools for visitor flow and resource management would function effectively in high-visitor attractions.

These Assumptions are not intended to describe empirical reality directly, but rather to structure the analysis by identifying where real-world operational practices align with or diverge from system-level expectations. In this way, the framework supports an examination of the practical conditions under which digital tools could realistically and responsibly support daily operations in high-visitor environments.

A total of six Assumptions were identified. The first five Assumptions primarily reflect functional conditions related to the feasibility and operational viability of digital tool implementation, while the sixth Assumption addresses sustainability as a non-functional consideration. This final Assumption was included to examine whether sustainability acts as an operational driver in practice, rather than remaining solely a policy-level aspiration.

The six Analytical Assumptions are defined as follows:

- ***Assumption 1*** – *Predictable patterns:* Visitor patterns are sufficiently predictable to support modeling and planning
- ***Assumption 2*** – *Data availability:* Structured operational data is available and of sufficient quality
- ***Assumption 3*** – *Staff capacity:* Staff have the capacity to adopt and use new digital tools
- ***Assumption 4*** – *Resource availability:* Financial and technical resources are available for digital implementation and maintenance
- ***Assumption 5*** – *Centralized control:* Operational authority is sufficiently centralized to act upon system outputs
- ***Assumption 6*** – *Sustainability driver:* Sustainability is treated as an operational driver rather than solely an aspirational or policy-level objective

These Assumptions form the basis of the Assumption Framework used to assess empirical findings across the participating venues. Each Assumption is evaluated according to the degree of alignment between the Assumption and the operational realities described by participants.

To support comparative analysis, *Figure 2* presents the evaluative framework used in this study. Assumptions are categorized using three evaluative labels: *Match, Partial,* and *Mismatch*. In cases categorized as *Partial*, an additional indicator may be applied: "+" to represent a tendency toward sufficient alignment, and "–" to represent a tendency toward insufficient alignment. This distinction acknowledges that qualitative data rarely yield binary results, and deepens the meaning of partial results. The following sections present the results of applying this framework across the studied venues.

Rating	Description
Match	Requirements are clearly present. A well-designed tool could operate here without major adaptation.
Partial +	Requirements partially present, leaning toward sufficient. Gaps exist but are managable with moderate effort.
Partial	Requirements partially present, neutral. Meaningful gaps that would require changes before a tool could operate reliably.
Partial -	Requirements theoretically present but functionally insufficient. The tool would face fundamental limitations, even with adaptation.
Mismatch	Requirements absent. A tool operating on this assumption would fail or produce misleading outputs in this context.
N/A	Assumption not applicable to this participant's role or venue type. Excluded from analysis for this case.

Figure 2: Rating Matrix

5.2 Assumption Matrix

Figure 3 maps all six Assumptions against the six venues included in the analysis. Art Museum (pre-study questionnaire) is excluded from the matrix due to the shallower data format, as noted in Section 3. Vasamuseet is included with N/A ratings due to participant's role and areas that it covers. It should be noted that the purpose of the matrix is not to compare venues against one another nor evaluate their operational effectiveness, but rather get an overview of how accurate preexisting assumptions are across HVAs. Additionally, the extended version of the Assumption Matrix, which includes detailed notes for each rating, can be found in Appendix A.1.

Assumptions	NRM Naturhistoriska Riksmuseet	SKM Scenkonst-museet	HM History Museum	IM Interactive Museum	EAV Edvard Andersons Växthus	VM Vasamuseet
Assumption 1 Predictable patters	Partial	Partial	Partial -	Partial +	Match	N/A
Assumption 2 Data availability	Partial	Partial	Partial -	Partial +	Partial -	Partial -
Assumption 3 Staff capacity	Partial -	Partial	Partial	Match	Mismatch	Partial -
Assumption 4 Resource availability	Mismatch	Partial -	Mismatch	Partial	N/A	Mismatch
Assumption 5 Centralized control	Mismatch	Partial	Mismatch	Partial +	Mismatch	Mismatch
Assumption 6 Sustainability driver	Partial +	Partial +	Partial -	Partial	Partial -	Partial

Figure 3: Heatmap/Assumption Matrix (Simplified)

5.3 Assumption Analysis

Having presented the Assumption Matrix in Section 5.2, this section examines each assumption individually. For each, the evidence from across the participating venues is assessed and a rating is assigned and explained. The analysis moves from the assumption with the strongest overall alignment (Assumption 1) to those with the most consistent mismatches (Assumptions 4 and 5), before addressing the sustainability assumption (Assumption 6) which occupies a distinct analytical position as the only non-functional condition in the framework.

5.3.1 *Assumption 1 - Predictable patterns*

All venues experienced some level of unpredictability, but frequency varied; this unpredictable demand was mainly due to the environment or human behaviour. There is only one match for this Assumption, Edvard Andersons Växthus, directly related to the venue's visitor volume and absent need for predictive tools. The four other venues get a range of partial ratings, based on size, demand, weather, events, venue type, and tools available. Naturhistoriska Riksmuseets (NRM)'s excessive visitor jumps add complexity to these patterns, going from 500 to 5000 visitors in a day, but staff remain on standby to respond to sudden increases in demand. The difficulty of foreseeing the ripple effects of operational decisions was also noted by the NRM participant: *"if you have complex operations, it's going to be much more complicated because the ripple effects of decisions or different paths that you want to take can be super difficult to predict [...] I think very few people understand how everything is connected."*. Similarly, Interactive Museum's volumes can jump to 16,000 only on Kulturnatt; however, it is planned for and other events are linked to the database, as well as used for trend forecasting. History Museum (HM) rarely has crowds, but as a newly established museum, there is no strong historical baseline for prediction, a reality

captured simply by HM's participant: *"when you're working and interacting with people, things happen."* Lastly, Scenkonstmuseet works with personas and segment data. Their core challenge is attracting new visitor segments, not managing the current ones.

Assumption 1 shows the highest overall ratings across all venues. This suggests that visitor pattern predictability is the condition most closely approximated in practice. However, the presence of only one full match indicates that even this condition is only partially satisfied across the dataset, meaning that ideal assumptions about predictability are not consistently met.

5.3.2 *Assumption 2 - Available and structured operational data*

Although data is present at all venues, it must be considered alongside experience; experience dominates real-time decisions at every venue. The existing data is mostly retrospective, not live, except for data used by max capacity alert systems at some venues. In some cases, data is fragmented across multiple systems and areas. Visitor counts, energy figures, staffing schedules, and sustainability reporting sit in separate systems with no shared data layer. Digital planning tools that require integrated data inputs across these domains would face a significant infrastructure gap across the majority of venues studied.

Most venues focus on visitor data, i.e., ticketing and volumes, as well as resources, primarily energy, since in most cases it is mandatory to be monitored and reported by the state or owner, mainly Statens Fastighetsverk (SFV). The degree of operational data tools used for collection ranges from simplified and mechanical, e.g., at cash register, to more sophisticated ways, e.g., a structured database used to identify trends and predictions. Additionally, an occasionally mentioned limitation for data collection, especially regarding visitors, is GDPR, which remains an important constraint to acknowledge.

Assumption 2 falls in the mid-range in terms of alignment across venues, with predominantly partial ratings. This indicates that data infrastructure exists, but not in a form that fully supports the requirements of digital planning tools. A consistent pattern across cases is that data is collected but not integrated, and primarily used retrospectively rather than as a real-time operational input. For tools relying on structured, accessible, and cross-domain data, this represents a fundamental limitation to effective implementation.

5.3.3 *Assumption 3 - Staff capacity to adopt new tools*

This Assumption showed the greatest variation across cases, ranging from a full match to a mismatch. Interactive Museum (IM) demonstrated strong openness and competency for adopting new digital tools, which may be influenced by the venue's interaction-driven nature. Additionally, a mix of technical and cultural staff drives adoption culture. On the other hand, Edvard Andersons Växthus (EAV) presents gaps mainly due to the time required to adopt a new tool, while age is also playing an important role. There is a noticeable spectrum of capacity and comfort level to adopt new digital tools.

AI was mentioned across most venues, either used individually or in early stages of exploration, e.g. text recognition, document reading, search, translation, and pattern recognition. Certainly, the previously mentioned GDPR remains a constraint to fully adopting these tools, even when individuals are capable. Another constraint for digital tools is that the staff's role determines their access to these, as mentioned by NRM: *"Can't use specialized stuff because not everyone will have a fancy laptop."* At the History Museum, the constraint is institutional rather than individual, as one participant stated, "it's a decision for the government." At Edvard Andersons Växthus, the challenge is more personal: *"there was a system I knew, but they changed it [...] I wasn't here due to sickness and missed training [...] it can be hard due to my age."*

Assumption 3 also falls in the mid-range, due to majority of partial ratings, but shows the widest variation across cases. The distribution from a full match (IM) to a mismatch (EAV) indicates that staff capacity for digital tool adoption is not uniform across the sector. Instead, it varies depending on organizational structure, institutional context, and local practices. This suggests that staff capacity cannot be assumed as a baseline condition and must be evaluated at the venue level prior to any digital implementation.

5.3.4 *Assumption 4 - Resource availability for digital investment*

This Assumption is one of the two with the most mismatches. The rating presents four mismatches out of six venues, with the two remaining being “Partial” and “Partial -”. There are no observed matches, and this is mainly due to budget constraints, as well as cycles, in control of the government. Another strongly present factor is the investment required to change digital tools, time, legal review, and effort, adding a layer of complexity beyond monetary cost. As one participant at NRM described, *“I need to apply for money... investment money, which is not part of the normal budget, it's actually an add-on... the museum borrows money from the state.”* The highest scoring venue highlighted that cost is a big part, but not the only factor, signaling that scalability and longevity are required. This pattern was echoed across venues, as one participant noted, *“it's quite a process to start using one new tool.”*

Assumption 4 is the lowest-ranked assumption overall, indicating a substantial gap in resource availability across all cases. This suggests that resource constraints alone would prevent sustainable tool adoption at the majority of venues studied regardless of other conditions.

5.3.5 *Assumption 5 - Centralized operational control*

This assumption is the second one with the most mismatches (four), with the remaining two ranking as “Partial” and “Partial +”. In most cases, it is a matter of having multiple stakeholders, split ownership at times, an example is NRM when discussing sustainability accountability with external operators: *“if they (restaurant) do something that is not very sustainable, we try to regulate that in our contracts, but [...] we can't necessarily tell them to do so. [...] It's their operation."* This significantly challenges the power to act on the outputs yielded by a digital tool, as coordination across multiple organizations or stakeholders requires time, alignment, and following an iterative approval cycle. Furthermore, split internal control poses another hurdle, especially when the information is scattered across areas. One participant at History Museum illustrated how structural fragmentation shapes operational authority: *“since we are part of a large organization, there's one department responsible for the buildings and energy use and [...] all those kinds of operations.”* At Vasamuseet, a participant described a similar dynamic in relation to sustainability responsibilities: *“that doesn't really work that way [...] we have specialists in different parts, and we have them (the state) that work with the properties [...] an example is that when we're renting, we should always aim to reduce energy consumption for heating. How? That's not our business; that's someone else."* Additionally, working with third parties and contractors can have both positive and limiting effects. Venues like Scenkonstmuseet take it as an opportunity to integrate sustainability in these practices, but it also poses a risk of having a negative or limited influence.

Interactive Museum, the closest match for Assumption 5, highlights team leaders on each shift who have clear operational authority and knowledge. In addition, the centrally managed system allows changes to be reflected throughout the venue and departments. Granting the possibility to also take over roles for the day, when a staff member is unable to. Having the same system makes it simpler to adapt and more understandable. As one participant explained: “*we try to work with systems that work for all of our things. [...] Having the same system in the information desk, as we have in the movie theater, as we have for school bookings, as we have in the restaurants. It helps a lot in managing short changes [...]. Everyone can do all the changes needed.”*

Assumption 5 shows a predominantly weak alignment, with the highest number of mismatches. This indicates that centralized decision-making authority is generally limited, with fragmented governance structures affecting the ability to act on outputs from digital systems.

5.3.6 *Assumption 6 - Sustainability as an operational driver*

All venues shared a high value, interest, and awareness towards sustainability, partially matching this Assumption. The studied venues have made changes in their daily operations, mainly towards single-use materials, often used by visitors. Moreover, when looking at sustainability in a more multifaceted way, there was more distinction among venues. The main gap is between venues where sustainability is structurally embedded and venues where sustainability is aspirational. As previously mentioned, sustainability is present in all cases; however, there is a strong correlation between the formal establishment of a sustainability structure and direct integration in a venue’s daily operations.

Priorities varied when participants were asked about operational decision-making. At NRM, sustainability was described as deeply embedded: *"sustainability is constantly measured on. It's very, very top of mind [...] it's very rare for us to say let's start from a blank slate and just buy new stuff."* At Scenkonstmuseet, sustainability is actively integrated into procurement: *"when we do tenders for employing consultants, [...] it's also a chance for us to [...] make them do good choices [...] that's more important than [...] cheap."* At the Interactive Museum, the presence of a dedicated sustainability coordinator was described as structurally relieving, allowing other staff to focus on their own responsibilities while sustainability risks are consistently flagged and addressed.

At other venues, sustainability competed more directly with other priorities. As one participant at the History Museum acknowledged, *"economy and visitor experience is a higher priority [...] the most important thing is to attract visitors in different ways."* At Edvard Andersons Växthus, sustainability and operational necessity were intertwined, when asked whether the automated climate control systems were driven by plant needs or environmental concern, the participant responded: *"both, I think, yes. And the cost also. [...] Without that (oil-based heating), it's better for the environment, and it's cheaper, of course."*

Vasamuseet offered the longest institutional perspective, noting that they were among the first museums in Scandinavia to obtain ISO 14001 certification and have been working with sustainability formally for many years. The participant also highlighted the scale of impact achievable through the broader Djurgården area, and noted that the organization is now shifting attention toward social sustainability, having largely addressed its environmental targets.

Assumption 6 is one of the stronger overall cases, with only partial ratings and no full matches or mismatches. This suggests that while sustainability is present across all venues, it is not consistently embedded as an operational driver, but rather remains at a strategic or aspirational level.

5.4 Synthesis

Having assessed each assumption individually in Section 5.3, this synthesis draws together what the matrix collectively reveals in relation to the research question. A full match across all assumptions would indicate that a venue already operates under the conditions digital planning tools require, meaning sustainable introduction would be feasible without significant prior adaptation. The absence of full matches across the dataset therefore carries direct analytical weight: it indicates that the conditions for sustainable digital tool introduction are currently absent or insufficient at the majority of venues studied.

For Assumptions 1-5, the operational Assumptions, partiality is present in pattern predictability, data availability, and staff capacity. However, resource availability and centralized control present the highest mismatches. Meaning that operational conditions are present but not consistently positive. The predominant barriers are structural, as primarily observed in A4 and A5; the constraint comes from institutional architecture rather than individual venue organization. This is most visible in Interactive Museum's profile, which approaches the most positive overall conditions and is also the most commercially oriented venue in the dataset.

In addition, Assumption 6, sustainability as operational driver, is unevenly integrated across all venues. It is present as an aspiration, but it only becomes operational where a dedicated sustainability function exists. Moreover, digital tools are utilized at all studied venues, but they serve as administrative instruments, not sustainability ones.

The presented findings indicate that the conditions for a sustainable tool introduction are currently absent or insufficient at the majority of HVAs in this study. These findings form the basis for the discussion that follows, which examines what this means for the sustainable introduction of digital tools in this context.

6 DISCUSSION

The analysis presented in Section 5 established what current practice reveals about the conditions for sustainable digital adoption. This section now asks what those findings mean, for the institutions themselves, for how sustainable introduction should be understood, and for the design of future tools. Section 6.1 addresses the conditions for responsible introduction,

connecting findings to Corporate Digital Responsibility principles. Section 6.2 introduces two contextual constraints specific to the HVA sector. Section 6.3 reflects on methodological considerations. Sections 6.4 and 6.5 discuss implications for High-Visitor Attractions and for tool design and future research respectively.

6.1 Responsible Introduction Conditions

A key finding has been the limited possibilities to make changes or decisions due to structural barriers. The most consistently present barriers were state governance, authority fragmentation, and outsourced infrastructure, conditions that directly limit both the monetary and organizational resources available for digital tool adoption. This pattern reflects the institutional architecture of publicly funded High-Visitor Attractions (HVAs) rather than individual organizational failure, it means responsible digital adoption cannot be achieved through venue-level action alone. The exception is Interactive Museum, whose more positive profile across most Assumptions correlates with its commercial orientation, suggesting that the structural barriers identified are specific to publicly funded HVAs rather than inherent to the sector as a whole. This raises a question that the current study cannot fully resolve: whether the structural barriers identified are a permanent feature of publicly funded HVAs or whether they can be addressed through institutional reform, policy change, or alternative funding models. If the conditions for digital adoption are structurally tied to commercial orientation, this would suggest that the sector's sustainability digitalization potential is fundamentally constrained for publicly funded venues, a finding with implications beyond individual organizations.

A second, equally significant constraint is the absence of a consistent sustainability mandate across venues. Where sustainability does not yet consistently drive decisions, tools designed to support sustainable operations lack the operational foundation to function. A tool cannot create the sustainability priority it requires, it can only serve one that already exists. Without a shift in how sustainability is prioritized at the institutional level, this foundation will remain absent in most venues. In Corporate Digital Responsibility (CDR) terms, responsible introduction requires that sustainability is already an operational driver, not only an aspiration. This structural picture also has implications for how tools themselves should be evaluated before introduction.

Tools introduced to support sustainable operations should be sustainable themselves and be evaluated against CDR principles prior to adoption. This includes assessing the environmental footprint of the digital infrastructure required (energy consumption, hardware lifecycle, and processing demands), as well as the organizational risks of introduction: rebound effects, shifted environmental burdens, and increased complexity. A tool that improves one dimension of sustainability while generating costs in another does not constitute responsible digitalization. In the HVA context, where operational capacity is already constrained, the organizational burden of maintaining and updating new digital systems is a particularly relevant consideration.

6.2 Contextual Constraints Specific to Studied High-Visitor Attractions

There are some contextual constraints observed, specific to the studied HVAs. Adding an extra layer of complexity, the analysis identifies two: Umbrella-Time Dilemma and the Sustainability-Heritage Quandary, each discussed below.

Included in this study were venues classified as umbrella organizations (organizations that oversee and coordinate a group of related entities under a shared governance structure) and more independent venues. Umbrella organizations often have split authority or a preestablished hierarchy with multiple decision checkpoints. In these situations, something that will be referred to in this report as the Umbrella-Time Dilemma arises, a term introduced by the author to describe a trade-off can emerge between organizational reach and implementation speed. An umbrella organization may enable a bigger impact all across with even a small change, but will require more effort and time due to its structure and complex decision-making processes. While an independent organization may be able to implement a change quicker due to its autonomy, but with possibly a smaller impact due to its more limited organizational reach. This distinction is considered in relation to organizational capacity, economic capacity is treated separately as it may vary on size and, where applicable, by the level of funding allocated by the state.

A tension between sustainability objectives and cultural heritage preservation was observed across all venues, each of which is responsible for safeguarding culturally or historically significant assets (e.g., the Vasa ship, museum collections, or botanical specimens from diverse ecosystems). In practice, cultural heritage requirements often take precedence over environmental sustainability and, in some cases, economic considerations. When cultural heritage preservation is at stake, it becomes the top priority for these venues, particularly in cases involving highly sensitive collections that depend on tightly controlled environmental conditions (such as temperature, humidity, or lighting). These constraints can limit the flexibility of operational changes aimed at improving sustainability performance. So the Sustainability-Heritage Quandary emerges, a term introduced by the author, choosing between these two will inevitably affect the other, but does it always have to be an unfavorable impact? Not necessarily, the coexistence of heritage preservation and environmental sustainability depends on how operational constraints are negotiated within each context. Importantly, this cannot be resolved through digitalization alone, as it reflects a deeper tension between the accessibility and preservation regulations that define these institutions, and the resource reduction logic of most environmental sustainability frameworks. Consequently, any digital tool introduced into this context must account for this, it cannot treat sustainability as an unconstrained optimization goal.

Digitalization may support more sustainable operations within heritage environments, but only when its application is situated within the specific operational and preservation requirements of each institution. On occasions, environmental benefits may even emerge as secondary effects of operational necessity, e.g., Edvard Andersons Växthus uses an automated environmental control system regulating temperature, ventilation, and lighting conditions within exhibition spaces. The system adjusts dynamically based on external environmental conditions, such as temperature and daylight levels, including mechanisms for automated window operation and lighting adjustment. Its primary driver is conservation requirements rather than sustainability optimization, however, it reduces energy use as a secondary effect.

6.3 Sustainability and Ethics

Sustainability is the central focus of this study, its three dimensions: environmental, economic, and social, are each present across the findings. Environmental sustainability is the primary focus, examined through resource management practices and the conditions under which digital tools could support sustainable energy, water, and waste use. Economic sustainability surfaces throughout the analysis, particularly in the structural barriers identified, where budget constraints, government funding cycles, and competing institutional priorities limit the scope for digital investment. Social sustainability, while not a primary focus, is implicitly present in the public-facing nature of HVAs, their role in cultural access and heritage preservation, and in the shift toward social sustainability priorities observed at Vasamuseet.

The thesis also engages with ethics at two levels. At the research level, participation was voluntary, informed consent was obtained from all participants, data was handled confidentially, and anonymization was applied where requested by participants or their institutions. In terms of content, Corporate Digital Responsibility is used throughout as a framework for the ethical dimensions of technology adoption in this context. The risks of rebound effects, shifted environmental burdens, and increased organizational complexity, each discussed in Section 2.2, represent ethical obligations for institutions considering digitalization. Any digital tool introduced into HVA operations should therefore be evaluated not only for its intended sustainability benefits, but for the unintended consequences it may introduce for staff, visitors, and the environment.

6.4 Method Critique and Study Considerations

The variation of roles for participants in this study produced enough information for the results and analysis. However, it is acknowledged that each role had a different level of expertise and access in the areas studied. From what has been observed, it is common for at least two different people to have operational knowledge and sustainability expertise. Retrospectively, the dataset could be richer if the study were done by pairing a sustainability expert and an operational expert. Due to access and time limitations, on both the researcher's and the venues' part, this was not possible.

Additionally, the geographical scope for this study, being entirely set in Stockholm, enhanced practicality, but it also presents a limitation for the findings since Stockholm’s context, culture, and governance structure can affect its outcomes. This can be the case for the structural barriers identified, such as landlord arrangements, budgeting, and loan processes,

which could differ from those in other Swedish cities or other countries. Regardless of geographical limitations, the dataset provides a diverse overview due to the variety of venues included. The participating organizations differed in structure, size, and thematic focus, which contributes to the robustness and transferability of the findings.

6.5 Implications for High-Visitor Attractions

The findings suggest that the effectiveness of digital tools in supporting sustainability within high-visitor attractions depends not only on technological capabilities, but on the degree to which sustainability is embedded within organizational structures, operational practices, and institutional priorities, a condition that this study found to be partially met at the majority of venues studied. Sustainability was acknowledged as an important objective; however, it was not always integrated into day-to-day operational decision-making or digital planning processes. This reflects broader discussions within software engineering and digital responsibility literature, where environmental sustainability is often underrepresented in requirements prioritization and treated as secondary to immediate operational concerns [28].

A key implication for museums and similar institutions is the need to integrate sustainability more directly into everyday operational processes. Rather than treating it as a separate strategic objective, sustainability should be embedded in routine decisions such as planning, procurement, and technology adoption processes. This also includes placing sustainability expectations on external stakeholders such as contractors, software providers, and infrastructure partners involved in digital implementation.

Second, the study highlights the importance of operational alignment when introducing digital tools. The findings demonstrate that digital systems rely on assumptions about the operational environment in which they are deployed. When these assumptions do not align with operational realities, digitalization efforts may create additional complexity or fail to produce meaningful sustainability improvements. As such, HVAs should evaluate digital tools in relation to their specific operational context rather than adopting them as universally applicable solutions.

The results also indicate that internal coordination structures play a critical role. Fragmented responsibilities and distributed decision-making can limit the ability to act on sustainability goals in practice. Strengthening internal alignment and clarifying responsibility for sustainability-related decisions may therefore improve operational implementation.

Finally, the findings suggest potential value in stronger collaboration and knowledge-sharing across museums and high-visitor attractions. Several of the identified limitations, particularly those related to staff capacity, sustainability expertise, and digital implementation, appear to extend beyond individual venues and may be influenced by broader institutional structures. Collaborative approaches, such as shared sustainability practices, common digital infrastructures, inter-organizational learning networks, or shared specialist roles, may help reduce duplicated effort and support more consistent approaches to sustainable operational management across the sector.

Together, these implications suggest that sustainable digitalization in high-visitor attractions depends less on technology alone and more on organizational alignment, institutional coordination, and context-aware implementation.

6.6 Implications for Tool Design & Future Research

There is a digitalization opportunity to improve sustainability in HVA daily operations, however, these will not be effectively introduced without first adapting the currently unmet assumptions. When designing future tools, the unmet Assumptions found in this study should be considered. The High-Visitor Attraction (HVA) sector, especially museums, has multiple limitations and complexity levels regarding digital tool introduction for sustainability improvement in daily operations. Future tools must be able to work with fragmented authority rather than assuming centralization, as well as accounting for the side-effects that this fragmentation causes, such as extended implementation cycles and multiple decision loops. Tools would also need to function with retrospective and fragmented data rather than requiring real-time integrated inputs, given that this type of information is currently limited across the sector.

Digital tools would need to support sustainability as an ongoing organizational process, not assume it already exists. Sustainability is present at HVAs but not yet embedded into daily operations. Furthermore, these venues' complexity must be considered, unlike a factory or an office building, a museum or greenhouse cannot simply minimize its environmental impact without potentially undermining the very reason it exists. Digital tools need to be designed with that constraint built in, not treated as a straightforward efficiency optimization problem. Additionally, digital tools that improve sustainability in daily operations should be sustainable themselves. When designing these tools CDR principles should be considered in the context where they will be introduced.

Looking ahead, several directions for future research emerge from the conditions identified in this study. The original direction of this study considered evaluating advanced digital tools already used in scientific and industrial sectors, such as digital twins, which have recently been applied in adjacent sectors including theme parks [30] and stadiums [31]. Their application in HVA operational contexts remains underexplored. However, given the identified research gap regarding operational conditions in High-Visitor Attractions (HVAs), the study was reoriented toward understanding those conditions first, a necessary precondition for any meaningful technology evaluation. Future research could build directly on the conditions identified in this study to evaluate whether, how, and which specific tools could be responsibly introduced into these environments, using the Assumption Framework developed here as a diagnostic instrument for assessing fit.

Although this study focused on daily operational impact in environmental sustainability, multidimensional sustainability should be acknowledged, without overlooking social and economic dimensions. Social and economic sustainability are discussed indirectly in this report, but worth examining in greater depth in future studies. Additionally, this study has answered the current state of HVAs and the conditions under which digital tools could be introduced to improve sustainability in daily operations, however, to what degree can this be done? Future research should also focus on analyzing digital tool potential for sustainability improvement, looking at quantitative data supported by the already collected qualitative data.

Lastly, it is recommended to study other types of HVAs in future research; due to accessibility, the majority of participants in this study were from museums. Looking at other types of venues can help identify their preparedness based on the preestablished assumptions, and if they differ from the ones observed in museums and learning-focused facilities.

In summary, considerable scope remains for further investigation, across sustainability dimensions, operational areas, quantitative data, specific technology assessment, venue types, and geographical contexts. This study has set a basis, direction, and next steps by identifying areas of focus and needs that must change before a digital tool can be introduced with the objective of improving sustainability in high-visitor attractions on their day-to-day operations.

7 CONCLUSION

This thesis addressed the research question: *how do high-visitor attractions currently manage visitor flow and resource use in daily operations, and what does this reveal about the conditions under which digital tools could support sustainable operations?* Empirically, the study found that HVAs manage these domains primarily through staff experience and judgment, with digital tools serving administrative rather than sustainability functions. Sustainability is present as an aspiration across all venues studied, but becomes operationally embedded only where a dedicated function exists. Furthermore, the institutional architecture within which these venues operate, state governance, fragmented authority, and outsourced infrastructure, constrains digital adoption more fundamentally than any individual organizational decision.

The study contributes an empirically grounded understanding of how HVAs currently operate and what this means for the responsible introduction of digital tools. This was achieved through the development and application of an Assumption Matrix Framework adapted to the HVA context, which revealed that the conditions digital tools require are absent or insufficient across the majority of venues studied. The analysis made visible four contextual factors that any future tool introduction must account for: the structural barriers embedded in public institutional architecture, the absence of a consistent sustainability mandate, the umbrella-time dilemma, and the heritage-sustainability quandary. These factors must be evaluated against CDR principles, ensuring that tools not only improve sustainability within operations but are sustainable

themselves, avoiding the rebound effects and shifted organizational burdens that irresponsible digitalization risks introducing.

The findings of this study are shaped by the venues and participants involved. The venue types, geographical context of Stockholm, and the roles of participants directly influence the results, and findings should be understood as indicative rather than generalizable. The exploratory design was appropriate for the research aims but leaves open questions that require further investigation.

Future research should prioritize the evaluation of specific digital tools for the HVA context, accounting for the structural and mandate conditions identified here. Extending the study to commercial HVAs, where structural barriers may differ significantly, would test whether the findings are specific to publicly funded venues or characteristic of the sector more broadly. Research into social and economic sustainability dimensions, which several participants described as growing priorities, would complement the environmental focus of this study. Finally, longitudinal research tracking how sustainability transitions from aspiration to operational driver, a process this study observed in different stages across venues, remains an open question that future research would be well positioned to address.

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A APPENDICES

In the appendix section, three levels of Appendix headings are available.

A.1 Extended Analysis

Figures 4 and *5* present the extended version of the Assumption Matrix introduced in Section 5.2.

Assumptions	NRM Naturhistoriska Riksmuseet	SKM Scenkonst-museet	HM History Museum	IM Interactive Museum	EAV Edvard Andersons Växthus	VM Vasamuseet
Assumption 1 Predictable patters	**Partial** 35k/week peaks vs 600 regular. Based on holidays. Adaptive to unexpected experiences, staff on hold.	**Partial** Personas + segment data used. Weather and day-of-week shape flow. Core challenge is attracting unknown audiences, not managing known ones.	**Partial -** New museum, no historical baseline. Only temporary exhibitions so far. Camera alarm mitigates peaks. Mostly no crowds.	**Partial +** Database links numbers to weather + events. Good trend forecasting. Kulturnatt outlier planned for.	**Match** Low volumes (~30–40k/year). "Not really" any hard situations. Rare exceptions for special events or flowering plants only.	**N/A**
Assumption 2 Data availability	**Partial** ViaMetric + ticketing used for forecasting. Energy monitored by SFV. Systems separated: Excel, task management tool, HyperGene.	**Partial** Entrance system tracks categories. Visitor study by Novus (periodic, external). "Mostly experience".	**Partial -** Digital visitor surveys. Energy reported monthly. Collections data is primary focus. No integrated planning data.	**Partial +** Database: visitor numbers, time, weather, events. Used to identify trends and predict staffing. Same information system across all parts of the museum.	**Partial -** Registration at cashier for statistics. Only adult/child/ school categories. No integrated system. Google Calendar for bookings.	**Partial -** Energy consumption measured and reported annually across 7 museums. Building owner (SFV) monitors daily.
Assumption 3 Staff capacity	**Partial -** 250 staff, mixed roles and literacy. Stakeholder diversity limits adoption.	**Partial** AI in early stages for search and translation. KTH collaboration ongoing. "Staff competence is a matter of education." No org-wide digital tool adoption described. Small team could be an advantage.	**Partial** AI integrated for surveys, text recognition, translations. However: government legislation requires everyone to know what they're doing. GDPR review for every tool.	**Match** "We have very competent co-workers, so the complexity of the system isn't a main part of it." AI-driven pattern recognition actively explored. Mix of technical and culture staff drives adoption culture.	**Mismatch** Some gaps due to missing training, and age. Can rely on colleagues who know the system.	**Partial -** ChatGPT used personally for document reading. Clearly individual, not organisational. State approval process for any new tool. No organization-level digital adoption discussed.

Figure 4: Extended Assumption Matrix (Part 1)

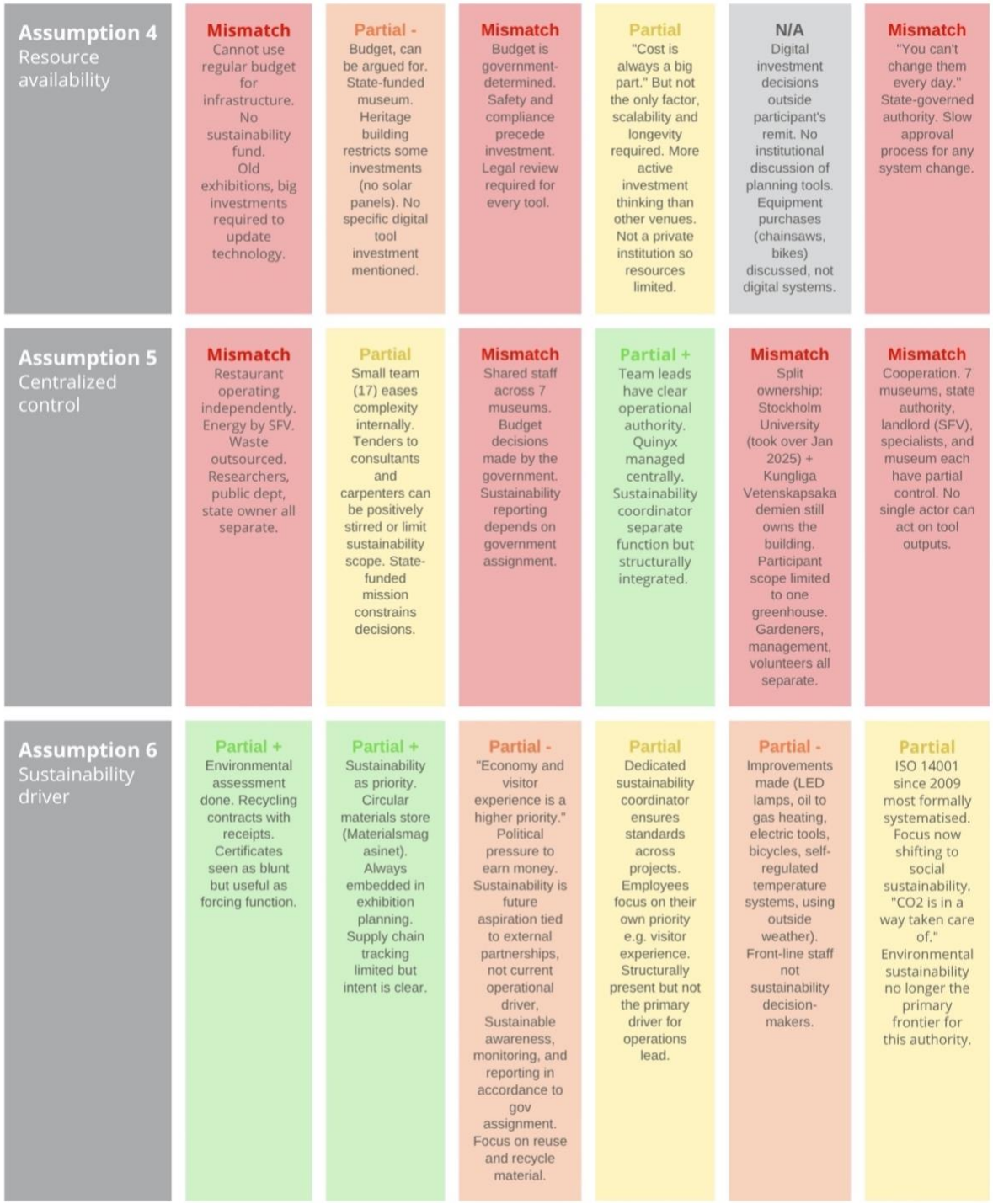


Figure 5: Extended Assumption Matrix (Part 2)

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